

NorthStar

Real-time evacuation intelligence
for crowds and critical facilities

TEAM NORTHSTAR

WHY NORTHSTAR

A blocked exit changes everything

4,000

people. Eight exits.
Uneven information.

Operators need to see rising pressure and adapt when a route fails.

Civilian safety

Stadium dispersal, public events and transport interchanges

Defense & security readiness

Facility evacuation, assembly points and continuity during disruption

One stadium. Two operating conditions

01 Post-event departure



Demand clusters at familiar gates

02 Earthquake response drill



Blocked exits shift queues elsewhere

25% less nominal exit capacity when 2 of 8 equal-capacity gates close

PROPOSED SOLUTION

A shared view. An explainable next action

Completed-system vision



Observe

People and available routes

Detect

Persistent gate pressure

Forecast

Near-term queue growth

Evaluate

Compare feasible interventions

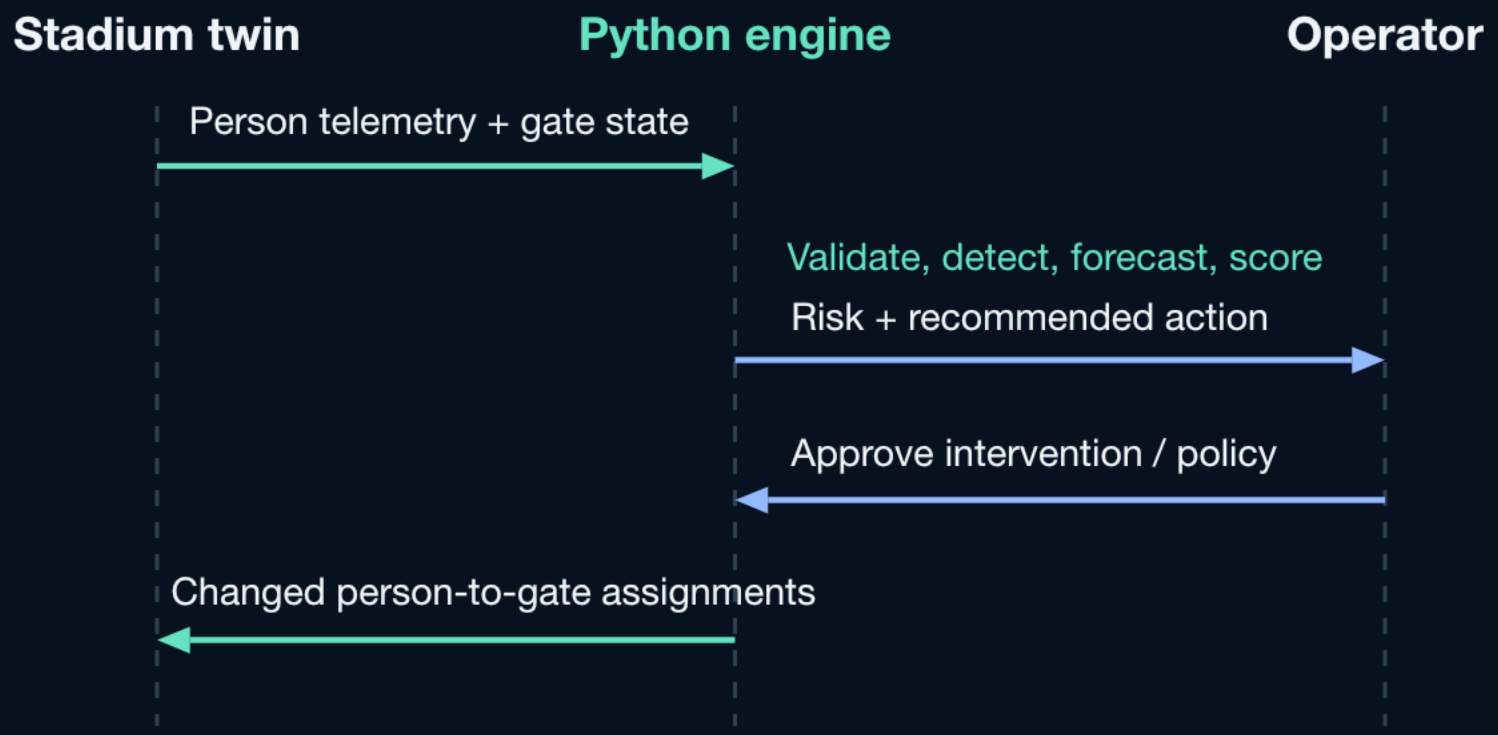
Reassign

Execute approved route changes

Commercial entry Paid venue pilot, then annual site software and calibration services

The technical decision loop

Target architecture: Python owns decisions. React executes and visualizes them.



Movement updates the next telemetry snapshot

React + TypeScript
Typed operator controls

Three.js + R3F
Interactive spatial twin

FastAPI + WebSockets
Continuous data exchange

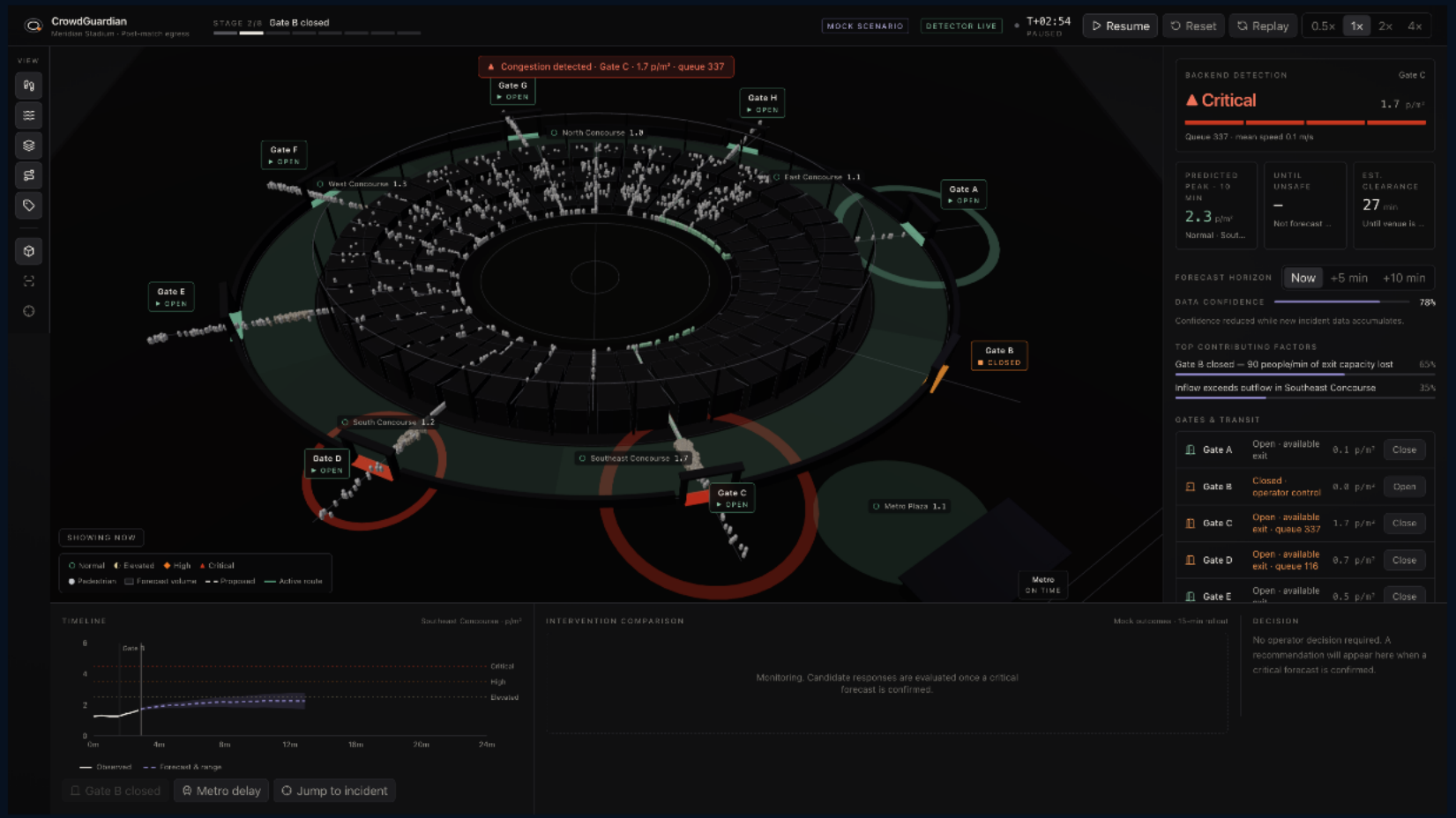
Deterministic algorithms
Inspectable decisions

Shared venue geometry
Consistent routing and detection

REAL PROTOTYPE DEMO

The stadium responds to a changing layout

Click the stadium to play the recorded demo



01 Normal departure

4,000 people use eight exits.



01 ALL OPEN



03 B / C / D CLOSED

02 Gate B closes

Python flags Gate C:
337 queued, 0.10 m/s.

03 B, C and D close

Demand shifts to other gates.

Actual app captures. Forecast and recommendation panels still use demonstration data.

A government pathway. A global product

01 Funding pathway

NSF America's Seed Fund

Up to $305k for Phase I feasibility R&D.
Project Pitch before full proposal.

AFWERX SBIR/STTR

Topic-aligned prototype contracts
for defense installation resilience.

02 Global deployment

California stadium
Shanghai campus
Indian event venue

Same engine. Local map, signage, language, rules and field validation.

Proposed U.S. pathways, subject to eligibility and matching topics. No awards or partnerships secured.

FUTURE SENSING AND PILOT GOALS

From simulated people to measured movement

Cameras, LiDAR and turnstiles → **Anonymous counts and calibrated tracks** → **Live venue state**

OpenCV for calibration and image processing

Detection + tracking for visible people. Density estimation for occluded crowds.

PILOT TARGET

15% less clearance time

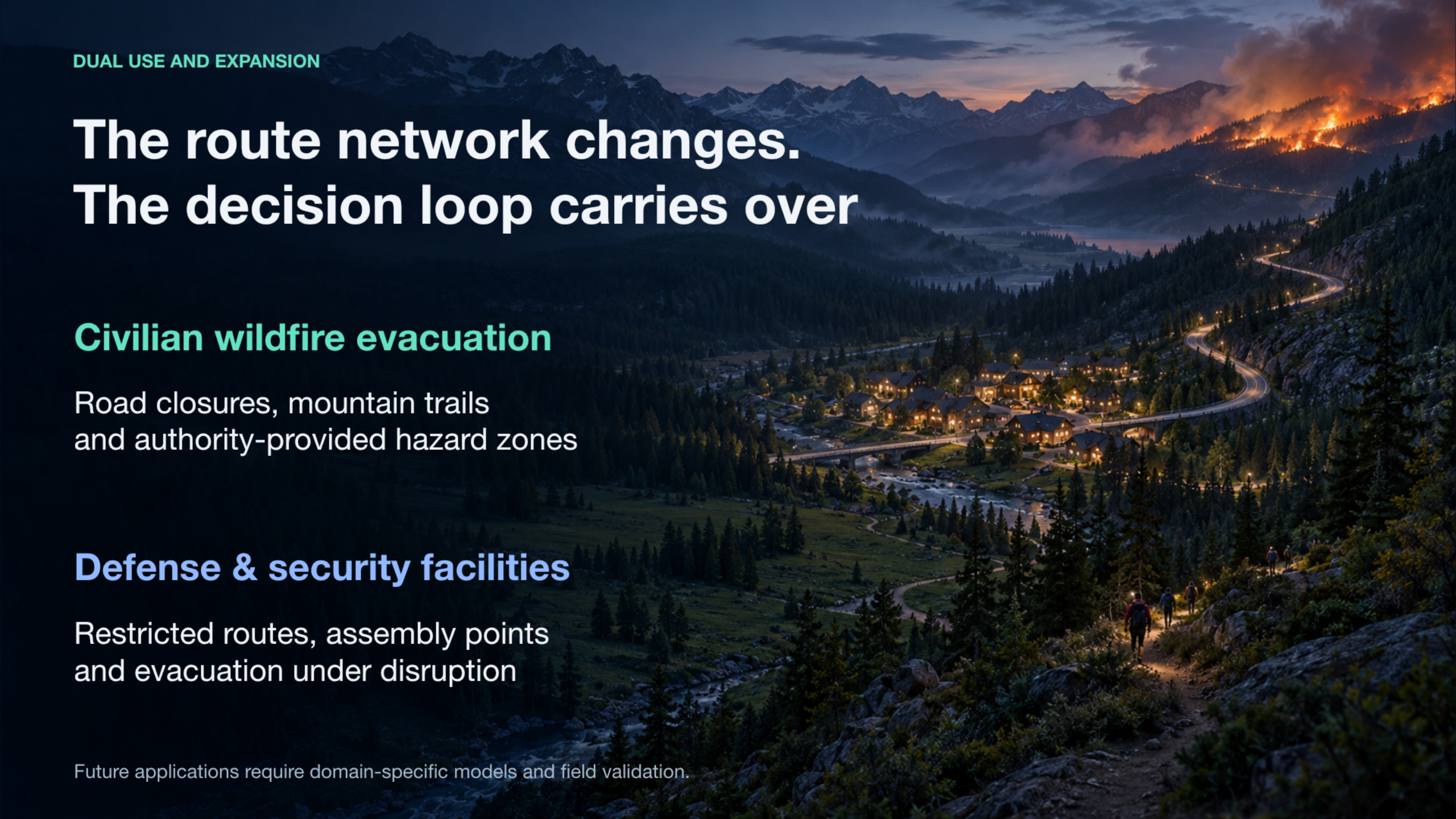
PILOT TARGET

20% lower peak queue

Unmeasured pilot targets against a fixed nearest-open-gate baseline across repeated runs.

DUAL USE AND EXPANSION

The route network changes. The decision loop carries over



Civilian wildfire evacuation

Road closures, mountain trails
and authority-provided hazard zones

Defense & security facilities

Restricted routes, assembly points
and evacuation under disruption

Future applications require domain-specific models and field validation.

Thank you

NorthStar

Seeking one venue partner for a six-week validation pilot

A calibrated model and controlled drills.
Measured outcomes.

Khushi Patel Ujjwal Puri

Mansi Surywanshi Aditya Lanjewar

